Sri Chaitanya IIT Academy.,India.

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A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT **Paper -2(Adv-2025-P2-Model)** **Date: 01-02-2026**

Time: 02.00Pm to 05.00Pm **GTA-13** **Max. Marks: 180**

KEY SHEET

MATHEMATICS

1	**A**	2	**B**	3	**B**	4	**A**	5	**AB**	6	**ABC**
7	**ABCD**	8	**CD**	9	**8**	10	**4.55-4.56**	11	**0.16**	12	**6**
13	**5**	14	**5**	15	**5**	16	**171**				

PHYSICS

17	**B**	18	**C**	19	**D**	20	**A**	21	**AC**	22	**ACD**
23	**AB**	24	**AD**	25	**5**	26	**330**	27	**14.77-14.78**	28	**2.40**
29	**1**	30	**2**	31	**15.70 - 15.80**	32	**11.50 - 11.60**				

CHEMISTRY

33	**A**	34	**B**	35	**C**	36	**A**	37	**ABD**	38	**ABD**
39	**BC**	40	**BD**	41	**5.7**	42	**25**	43	**1.66**	44	**4**
45	**12.5**	46	**48.25**	47	**10**	48	**0.15**				

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SOLUTIONS

MATHEMATICS

1. $$V^2=\frac{1}{36}\left[\vec{a}\,\vec{b}\,\vec{c}\right]^2=\frac{1}{36}\begin{vmatrix}\vec{a}.\vec{a} & \vec{a}.\vec{b} & \vec{a}.\vec{c}\\ \vec{b}.\vec{a} & \vec{b}.\vec{b} & \vec{b}.\vec{c}\\ \vec{c}.\vec{a} & \vec{c}.\vec{b} & \vec{c}.\vec{c}\end{vmatrix}=\frac{1}{12}\sqrt{3\sqrt{3}-5}$$

2. $$S_r=\frac{r^5+3r^4+2r^3+2}{r(r+1)(r+2)}=\frac{r^3(r+1)(r+2)+2}{r(r+1)(r+2)}$$

$$S_r=r^2+\frac{2}{r(r+1)(r+2)}$$

$$\sum_{r=1}^{10}S_r=\sum_{r=1}^{10}r^2+\sum_{r=1}^{10}\frac{2}{r(r+1)(r+2)}$$

$$=385+\sum_{r=1}^{10}\frac{1}{r(r+1)}-\frac{1}{(r+1)(r+2)}$$

$$=385+\frac{65}{132}$$

$$\left[\sum_{r=1}^{10}S_r\right]=385$$

3. $$J-I=\int\left(\frac{x+y}{xy}dy-\frac{x+y}{x^2}dx\right)$$

$$=\int\left(\frac{x+y}{x^2y}\right)(xdy-ydx)$$

$$=\int\left(\frac{x+y}{y}\right)\left(\frac{xdy-ydx}{x^2}\right)$$

$$=\int\left(\frac{x}{y}+1\right)d\left(\frac{y}{x}\right)$$

$$g(x)=\frac{y}{x}+l\,n\left(\frac{y}{x}\right)+c$$

Put $x=1\Rightarrow 1=1+c\Rightarrow c=0$

$$\therefore g(x)=\frac{y}{x}+l\,n\left(\frac{y}{x}\right)$$

g(e) = e + 1

4. Let the radius of the smallest circle be a. We find that the radius of the largest circle is 4-a and the radius of the second largest circle is 3-a. Thus, $4-a+3-a=5\Leftrightarrow a=1$. The radii of the other circles are 3 and 2. The sum of their areas is

$\pi+9\pi+4\pi=14\pi\Leftrightarrow(E)$

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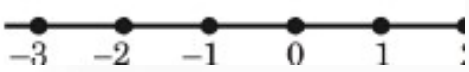
5. $(BC)_{21}^{r} = BC_{12} = \dfrac{2a-2}{a+2+2a^2} = -1$

6. Putting $2x = \sin\theta,$ we get

$$f(x) = \begin{cases} \dfrac{\pi}{6} - \sin^{-1}(2x,) & x \in \left(\dfrac{-\sqrt{3}}{4}, \dfrac{1}{2}\right] \\ \dfrac{-5\pi}{6} - \sin^{-1}(2x), & x \in \left[\dfrac{-1}{2}, \dfrac{-\sqrt{3}}{4}\right) \end{cases}$$

7. $H'(x) = (x^2-1)(x^2-4)(x^2-9)2x$

H' is odd $\Rightarrow H''$ is even and H''' is odd



8. $S_n = 1 + 22 + 33 + \ldots\underbrace{999\ldots9}_{9\ times}$

$T_n = \underbrace{nnn\ldots n}_{n\ times}$

$= n\dfrac{\left(10^n - 1\right)}{9}$

$T_n = S_n - S_{n-1} = \dfrac{n\left(10^n - 1\right)}{9}$

Also $S_3 = 356$

9. Let $\vec{a} = 2\hat{i} + 3\hat{j} + 6\hat{k}$

$\vec{b} = (\sin x \sin y)\hat{i} + \cos y\,\hat{j} + (\cos x \sin y)\hat{k}$

And $\vec{a}.\vec{b} = |\vec{a}||\vec{b}| \Rightarrow \vec{a} \| \vec{b}$

$\dfrac{\sin x \sin y}{2} = \dfrac{\cos y}{3} = \dfrac{\cos x \sin y}{6}$

$\tan x = \dfrac{1}{3}, \tan y = \dfrac{2}{3}\cos ecx$

$= \dfrac{2}{3}.\dfrac{\sqrt{10}}{1}$

$\tan^2 x + \tan^2 y = \dfrac{1}{9} + \dfrac{40}{9} = \dfrac{41}{9} = 4.55$

10. Let $z_1 = e^{i\frac{2m\pi}{101}}, z_2 = e^{i\frac{2k\pi}{100}}$

$\left| e^{i\frac{2m\pi}{101}} + e^{i\frac{2k\pi}{101}} \right| \geq \sqrt{2+\sqrt{3}}$

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$$\Rightarrow \sqrt{\left(2+2\cos\frac{2\pi}{101}(m-k)\right)} \geq \sqrt{2+\sqrt{3}}$$

$$\Rightarrow \cos\left(\frac{2\pi(m-k)}{101}\right) \geq \frac{\sqrt{3}}{2}$$

$$\left|\frac{2\pi(m-k)}{101}\right| \leq \frac{\pi}{6}$$

$$|m-k| \leq \frac{101}{12} \Rightarrow |m-k| \leq 8$$

$$P = \frac{101.16}{101.100} = \frac{4}{25}$$

11. $dy = \frac{(1-x)}{e^x}dx$ Integrate

$$y = xe^{-x} + c, f(0) = 0 \Rightarrow c = 0$$

$$y = xe^{-x}$$

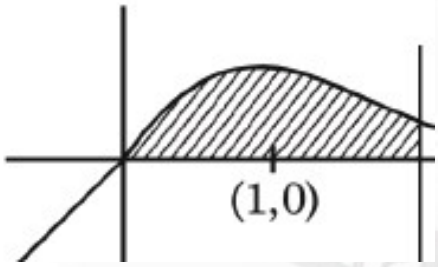
$$\frac{d^2y}{dx^2} = (x-2)e^{-x}$$

Point of inflection $= \left(2, \frac{2}{e^2}\right)$

$$A = \int_0^2 xe^{-x}dx$$

$$A = 1 - \frac{3}{e^2}$$

$$p+q = 5$$



12. $(x-c)^2 + (y-c)^2 = c^2, c > 0$

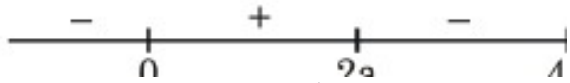
P = r, $\left|\frac{7c-12}{5}\right| = c \Rightarrow c = 1, 6$

Sum = 7

13. $f'(x) = \frac{(x-2a)}{x(x-4a)\log a} \geq 0$

$C-1$: $a \in (1,\infty) \Rightarrow \frac{(x-2a)}{x(x-4a)} \geq 0$

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$x \in (0,2a)$ so always ↑

C-2 : $a \in (0,1)$

$$\frac{(x-2a)}{x(x-4a)} < 0$$

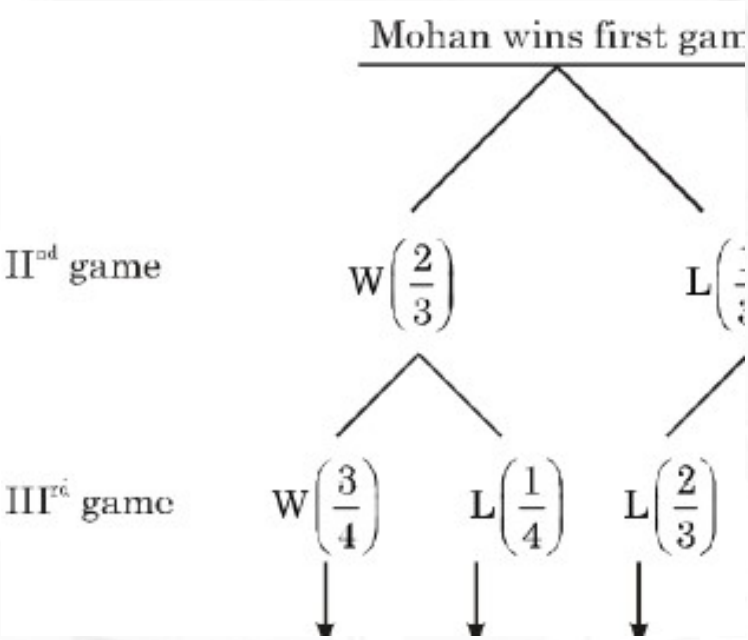


$x \in (2a,4a)$

$2a \le \frac{3}{2} \Rightarrow a \le \frac{3}{4} and\ 4a > 2 \Rightarrow a > \frac{1}{2}$

So $a \in \left(\frac{1}{2},\frac{3}{4}\right] \cup [1,\infty)$

14.

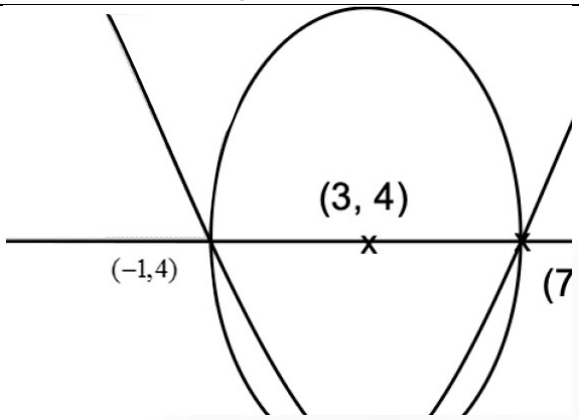


∴ Prob (mohan win 4th game) =

$$\frac{2}{3}\times\frac{3}{4}\times\frac{4}{5}+\frac{2}{3}\times\frac{1}{4}\times\frac{1}{3}+\frac{1}{3}\times\frac{2}{3}\times\frac{1}{4}+\frac{1}{3}\times\frac{1}{3}\times\frac{2}{3}=\frac{79}{135}$$

15. Given ellipse $\frac{(x-3)^2}{4^2}+\frac{(y-4)^2}{7^2}=1$ (vertical ellipse)

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Parabola can be taken as

$(x-3)^2 = A(y+3)$

It passes through (-1, 4)

$\Rightarrow 16 = 7A \Rightarrow A = 16/7$

$\therefore$ parabola is $7(x-3)^2 = 16y + 48$

$\therefore 16y = 7(x-3)^2 - 48$

$\therefore A = 7,\ H = 3, K = 48$

$\therefore \frac{A}{7} = \frac{H}{3} + \frac{K}{16} = 5$

16. Maximum number of points of intersection possible if there are no constraints $= {}^{20}C_2 = 190$. But 5 of these 20 lines are parallel. There will not be any intersection point from these 5 lines.

$\therefore$ Maximum number of points of intersection $= 190 - {}^{5}C_2 = 180$. Also the times $L_1, L_5, L_9, L_{13}, L_{17}$ pass through one point.

$\therefore$ We need to replace ${}^{5}C_2 (= 10)$ points by 1.

$\therefore$ The maximum number of points of intersection – 190 – 10 – 10 + 1 = 171

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PHYSICS

17. Fission of a nucleus of feasible only if the binding energy of daughter nuclei is more than the parent nucleus.
A = 70 will have same BE as 110 but A = 40 will have more B.E.
A = 100 will have same BE as 110 but A = 10 will have lesser B.E.
A = 90 will have same BE as 110 but A = 20 will have lesser B.E.

18. $k_{Bl}^{x}\frac{\pi\left(b^2-a^2\right)\Delta T}{l}=k_m\frac{\pi a^2\Delta T}{l}$

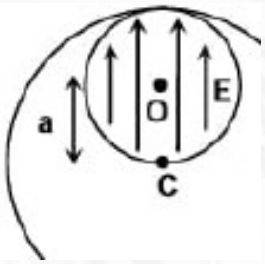
$\Rightarrow\frac{b^2}{a^2}-1=\frac{1}{3}\frac{b}{a}=\frac{2}{\sqrt{3}}$

19. Conceptual

20. Let the horizontal component of the velocity of the projected ball be u_0 and the vertical component v_0. The ball reaches the wall at point B after time, $t_0=\frac{v_0}{g}$, $H=\frac{v_0^2}{2g}$

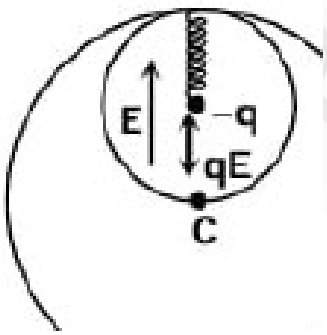
$1=e+2e^2+2e^3.....\propto=e+\frac{2e^2}{1-e}\qquad e=\sqrt{2}-1$

21. E in cavity is uniform and it is equal to $E=\frac{\rho a}{3\varepsilon_0}$



Time period of spring block system is independent of constant force so $T=2\pi\sqrt{\frac{m}{k}}$

Work energy theorem $qEX_{\max}-\frac{1}{2}kx_{\max}^2=0$



$X_{\max}-\frac{2qE}{k}=\frac{2q}{k}\frac{\rho a}{3\varepsilon_0}\qquad X_{\max}\frac{2\rho aq}{3\varepsilon_0 k}$

22. Conceptual

23. At

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$t = \frac{\tau}{3}.S = \frac{2a}{3} \therefore r = \frac{2a}{3}$

24. $K_{\max imum} = \frac{1}{2}\int_0^l (\lambda dx)\omega^2 (A\sin kx)^2$

So, $E_k = \frac{1}{4}mA^2\omega^2$

Also $< E_k >= \frac{E_k}{2} = \frac{1}{8}mA^2\omega^2$

25. $S_1P - S_2P = \frac{d^2}{2D} = \frac{2\times10^{-3}\times2\times0^{-3}}{2\times\frac{8}{5}} = \frac{5}{2}\lambda$

$(\lambda = 500nm)$

So when S is at ∞ there is Ist minima and when S is at S_2 there is last minima because d/1 = 4000 So the number of minima's will be 4001 and number of maxima's will be 4000.

26. Intensity of the source at the cross-section A

27. Conceptual

28. $\left(\pi r^2\right)\sqrt{2gy} = \pi X^2\left(-\frac{dy}{dt}\right)$

$\Rightarrow (r^2)\sqrt{2gy} = x^2\lambda$

$\Rightarrow y\alpha x^4$

$\Rightarrow n = 4$

29. $\Rightarrow F = \frac{W}{2\pi}\frac{\Delta h}{\Delta r}$

$F = \frac{W}{2\pi}\frac{r}{h} = 1$

30. Conceptual

31. Conceptual

32. Conceptual

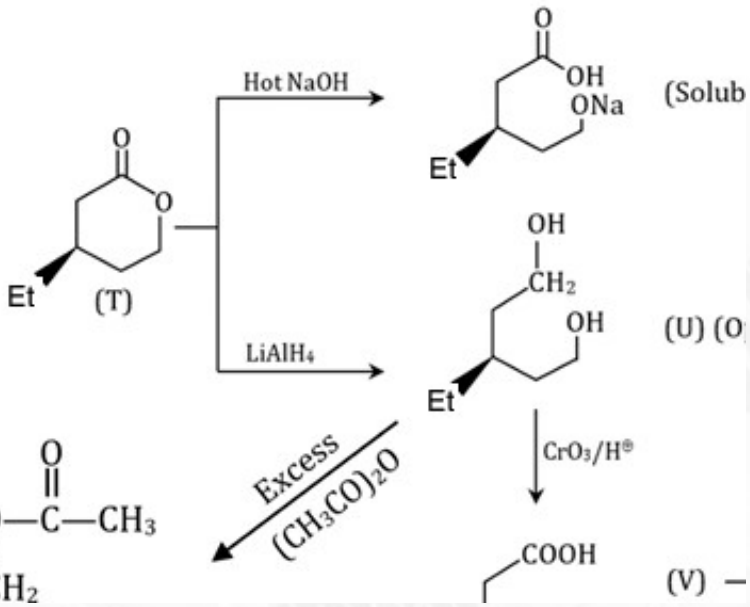
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CHEMISTRY

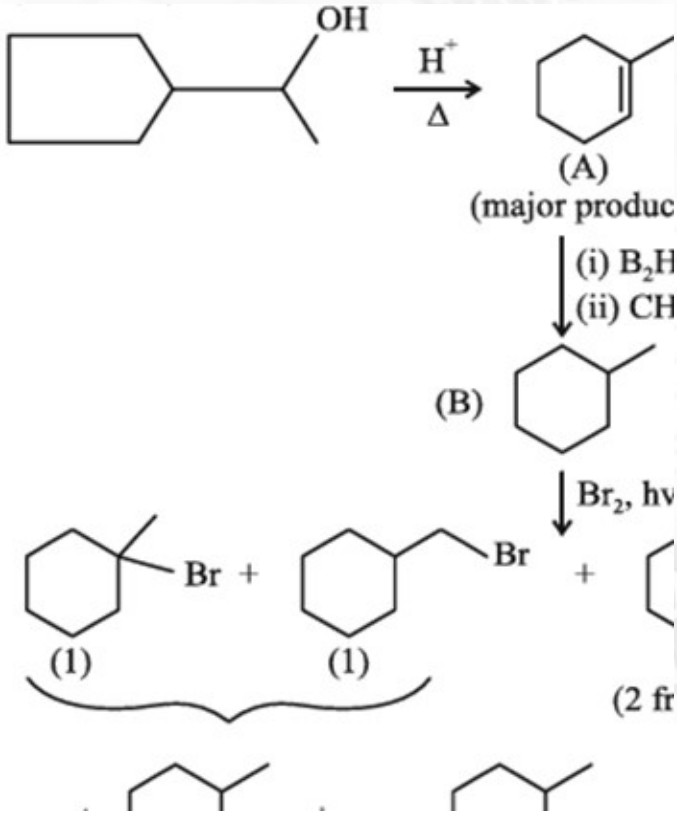
33. $Co(SCN)_6^{3-}$ has least spitting energy among the given

34. Cadmiam complex is less stable that gives $cds\downarrow$ $(yellow)$

35.



36.



Total 12 mono-bromo products are formed.

37. $U \propto \frac{1}{r^6}$ (rotating dipoles)

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$U \propto \frac{1}{r^3}$ (stationary dipoles)

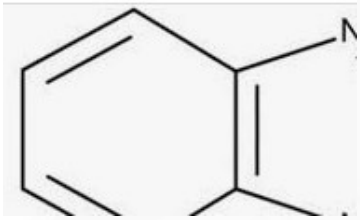
38. A) $P_4 \xrightarrow{OH^-} PH_3$

B) $SO_3^{2-} \xrightarrow{OH^+} SO_2$

C) $PCl_3 \xrightarrow{H_2O} H_3PO_3$

D) $NCl_3 \xrightarrow{H_2O} NH_3 + HOCl$

39.



(75% yield)
Benzotriazole
It is internal coupling reaction
D. U = 6

40. Only (vi) gives the test

41. $P^{OH} = P^{kb} - y\frac{0.2}{0.1} = 5.7$

$\left(OH^-\right) = 2\times10^{-6}$

$\left(Mg^{+2}\right)\left(4\times10^{-12}\right) = 1.2\times10^{-12}$

$\left(Mg^{+2}\right) = 0.3M$

M_{MgCl_2} in 200 ml $= 0.3\times0.2\times95 = 5.70$

42. The gas fallows PV^{-1} = constant

43. $MCl \rightleftharpoons M^+ + Cl^-$

$(a+b)(a) = 2.4\times10^{-11}$

$(a+b)(b) = 4.0\times10^{-11}$

$\Rightarrow (a+b) = (M^+) = 8\times10^{-6}$

$E_{cell} = \frac{0.06}{1}\log\frac{(0.1\times2)}{8\times10^{-6}}$

$= 0.264 \Rightarrow 0.26v$

44. $T_B = 2T_c;\quad T_D = 4T_c$

$(\Delta s)_{adia} = nR\ln\frac{V_2}{V_1} + nC_v l\,n\frac{T_2}{T_1} = 0$

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45. $1m = \frac{171}{342} \times \frac{1000}{m_{solvent}}$

Mass of solvent = 500 gm

$\therefore$ After ice is given out water left = 400 gms

$\therefore\ m = \frac{171}{342} \times \frac{1000}{400} = 1.25$

$\Delta T_f = 5 \times 1.25 \times 2 = 12.5$

46. $2HCOOK + 2H_2O \longrightarrow \underset{cathode}{H_2} + 2CO_2 + \underset{anode}{H_2} + 2\underset{aq}{KOH}$

$2F \Rightarrow 2 moles\ of\ H_2$

0.3 mole H_2 needed

$2 mole \Rightarrow 2F$

$0.3 \Rightarrow 0.3F$

$+ = \frac{0.3 \times 96500}{10 \times 60} \min = 48.25 MIN$

47. $Co(NH_3)_3(NO_2)_3 : f\ Am \Rightarrow 2$

$Co(NH_3)_3(ONO)_3 : f\ Am \Rightarrow 2$

$Co(NH_3)_3(ONO)_2\ f, + m(cAt) = 3$

$Co(NH_3)_3(NO)_2\ (ONO)\ f + m(cAt) = 3$

$\Rightarrow 10$

48 $\Delta x . \Delta v = \frac{h}{4\pi m}$

$\Delta x . \frac{\Delta v}{v} = \frac{h}{4\pi m x}$

$\frac{1}{4\pi} \times \frac{15}{100} = \frac{\Delta v}{4\pi}$

$\lambda = 0.15 A^o$

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